



IBM Professional Certification Program

Study Guide Series

Exam C1000-185

IBM watsonx Generative AI Engineer v1 -
Associate

Purpose of Exam Objectives

When an exam is developed, Subject Matter Experts work together to define the role the certified individual will fill. They define the tasks and knowledge that an individual would need to successfully perform this job role for the product or solution. This creates the foundation for the objectives and measurement criteria, which form the basis of the certification exam. Question writers then use these objectives to develop exam questions.

It is recommended that you review these objectives and ask yourself the following questions:

- Do you know how to complete the task in the objective?
- Do you know why that task needs to be done?
- Do you know what will happen if you do it incorrectly?

If you are not familiar with a task, go through the objective, perform that task in your own environment and read more information on the task. Use the Study Resource reference links below and search for the topic of the task. Review the documentation that you find until you *can* answer 'yes' to the above questions. If there is an objective on a task, there is a high likelihood that you *WILL* see a question about it on the actual exam. Review the recommended learning designed to prepare you to take the certification exam.

After reviewing the objectives in this guide and completing your own research, take the assessment exam. While the assessment exam does not indicate which specific questions were answered incorrectly, it does indicate overall performance by section. This is a good indicator of preparedness or if further preparation is warranted.

Study Resources

Below is a high-level list of resources to help when you are preparing for the certification exam. This list is not exhaustive and meant to help you should you need more information on topics listed below in the Study Guide. Use the high-level reference links and search for the topic of the task.

Detailed Exam Objectives

Section 1: Analyze and Design a Generative AI Solution

This section accounts for approximately 15% of the exam.

1.1. Understand the 5 Capabilities of GenAI models/LLMs

- Define and describe the five key capabilities of generative AI and large language models (LLMs)
 - Summarization
 - Classification
 - Generation
 - Code
 - Translation
 - Extraction
 - Q&A
- Review case studies or examples demonstrating each capability.
- Evaluate the impact of these capabilities in various industries
- Discuss potential future developments in LLM capabilities.

1.2. Articulate the Components in Gen AI Patterns

- Identify common patterns used in generative AI solutions.
 - Mixture of experts (MoE)
 - Variation Auto Encoders (VaE)
 - Transformer based models
 - Reasoning models
- Describe the components that constitute these patterns (e.g., input data processing, model training, output generation).
- Analyze real-world examples to illustrate how these patterns are applied.
- Create a diagrammatic representation of various generative AI patterns.

1.3. Understand the Limitations of GenAI/LLMs

- List the technical and ethical limitations of generative AI and LLMs.
- Discuss scenarios where generative AI may produce biased or incorrect outputs.
- Explore strategies to mitigate these limitations in practical applications.
- Evaluate the risks associated with the deployment of generative AI in sensitive or critical applications.

1.4. Understand Use Case and Identify Gen AI Application Opportunities

- Study various industry sectors to identify potential use cases for generative AI.
- Conduct needs analysis to determine how generative AI can address specific business problems.

- Propose generative AI solutions for identified use cases.
- Prepare feasibility reports for the proposed solutions.

1.5. Understand How to Choose the Appropriate Model for a Use Case

- Analyze the requirements of a given use case.
- Determine criteria for selecting an appropriate model based on performance, efficiency, cost, and ethical considerations.
 - Model selection based on Parameter size
 - Chat vs instruct
 - IBM Granite models
 - Billing classes
- Simulate decision-making processes to select the optimal model for a use case.

1.6. Articulate the Optimal Model Architecture Based on Use Case

- Define model architecture and its significance in AI solutions.
- Examine various model architectures and their components.
- Match model architectures with specific use cases based on technical requirements.
- Agentic architectures

1.7. Identify and apply various tools and techniques like AI agents, RAG , LangChain , etc.

- Understand the RAG Pattern
 - Introduction to the RAG pattern to Basic Search
 - Demonstrate the RAG pattern with LangChain
- AI agents

1.8. Understand security risks associated with LLMs, prompt engineering, prompt, and data

- Understand the Security Risks Associated with Inputs
 - Data Bias
 - Data Poisoning
 - Data Curation and Downstream Retraining
 - Data Privacy
 - Prompt Injection
 - Prompt Leaking
- Understand the Security Risks Associated with Output
 - Output and Decision Bias
 - Revealing Confidential or Personal Information
 - Toxic Output
 - Spreading Miss disinformation and toxicity
 - Harmful code generation
- Understand the Security and Privacy for Foundation Models

- Guardian models

Section 2: Prompt Engineering

This section accounts for approximately 16% of the exam.

2.1. Differentiate between zero-shot and few-shot prompting

- Introduction to Zero-shot prompting
- Introduction to few-shot prompting
- Try Zero-shot prompt sample
- Try Few-shot prompt sample

2.2. Design Prompts based on use case

- Choose right model for a use case.
- Try to converse with model for a chat use case.
- Try to converse with model to translate from one language to another.

2.3. Generate Prompt Templates

- Introduction to Prompt templates
- Evaluate Prompt Templates
- Creating Environment Templates
- Deploying Prompt Templates
- Tracking Prompt Templates

2.4. Determine the best model parameters for each GenAI prompt

- Introduction to model parameters
 - Decoding
 - Greedy Decoding
 - Sampling Decoding
- Understanding the Random Seed
- Understanding the Repetition Penalty
- Understanding the Stopping Criteria
- Understanding the Stop Sequences
- Understanding the Minimum and Maximum Tokens

2.5. Describe the benefits of using prompt variables

- Articulate what a prompt variable is
- Identify optimal static prompt text to replace with variable
- Summarize the value of reusable prompts

2.6. Describe the benefits of Prompt Lab

- Articulate the prompt editing options

- Chat
 - Structured
 - Freeform
- Demonstrate how to build reusable prompts
- Describe example input prompts and their use

2.7. Controlling model parameters

- Articulate the decoding process at a high level
 - Greedy decoding
 - Sampling decoding
- Describe the sampling decoding parameters
- Temperature
 - Top K
 - Top P
 - Random Seed
 - Repetition penalty
- Articulate stopping criteria and examples
 - Stop sequences
 - Minimum and Maximum tokens
 - model generation time limit

2.8. Articulate model risks

- Describe hallucinations in model output
 - Underlying causes
 - Techniques for avoiding
- Articulate the risks associated with personal information
 - Identify Personal information
 - Techniques for excluding
 - PII filter
- Hate speech, abuse, and profanity
 - Techniques for reducing risk
 - HAP filter
- Bias
 - Underlying cause of bias in outputs
 - Techniques for reducing bias
- Debate the risks of data bias and poisoning

Section 3: Fine-tuning

This section accounts for approximately 31% of the exam.

3.1. Understand the difference between hard and soft prompts

- Differentiate between hard prompts and soft prompts
 - Designed by humans or AI
 - Readability of the prompt
 - Explainability
- Articulate how soft prompts are generated
- Debate the benefits and drawbacks of soft prompts
 - Performance
 - Simplicity
 - Interpretability
 - Explainability

3.2. Reconstruct prompts to reduce the cost of using GenAI models

- Manage the token usage of each prompt template and model
- Detect inefficient prompt techniques
- Design cost-effective prompt templates
- Employ model parameters to reduce the generation cost
 - Stop sequences
 - Min/max token limits

3.3. Plan for Data elements for application usage

- Add data to a watsonx.ai project for tuning
- Inspect and validate data elements using Data Refinery

3.4. Articulate model quantization techniques

- Understand quantization is in the context of LLMs
 - Tradeoffs
 - Reduce precision
 - Reduce computational costs
 - Techniques

3.5 LoRA

3.6 Prepare the dataset for training

- Summarize the purpose of taxonomy tree -based curation
- Generate synthetic data using InstructLab
 - Describe LAB (Large-scale Alignment for chatBots) methodology

3.7. Customize LLMs with InstructLab

- Describe what the components of InstructLab
 - Taxonomy driven data curation

- Large scale synthetic data generation
- Iterative, large scale alignment tuning
 - Knowledge tuning
 - Skill tuning
- Articulate the InstructLab workflow

3.8. Generate synthetic data using the User Interface

- Describe the two options supported
 - Leverage your existing data
 - Create from your custom data schema
- Understand limitations on importing existing data sources and size limitations.
- Understand anonymization of imported data
- Understand the two algorithms for mimicking existing data
 - Kolmogorov-Smirnov
 - Anderson-Darling
- Understand differential privacy concepts and settings
 - Privacy budget
 - Privacy leakage probability
 - Random seed
- Understand sizing requirements for synthetic data generator

Section 4: Retrieval-Augmented Generation (RAG)

This section accounts for approximately 17% of the exam.

4.1. Describe what embeddings are in Context of GenAI

- Understand concepts of text embeddings.
- Describe different embedding models
 - IBM Embedding models
 - Third party embedding models

4.2. Generate vector embeddings utilizing models

- Perform converting text to embedding vectors
- Describe prerequisites for embedding API
 - Credentials
 - Project ID
 - Deployment ID
- Understand the choices and capabilities of vector databases
 - Purpose built vector databases
 - Vector extensions to popular databases

4.3. Describe when to use a vector database

- Understand the concept of a retriever
- Describe different types of retrievers
 - Vector databases
 - Embedded
 - Static
 - Watson Discovery
 - GitHub code retrieval API
 - watsonx Discovery
- Understand the capabilities of retrievers
- Understand use case driven selection of a retriever.

4.4. Develop using libraries and tools

- Understand the RAG Pattern
- Understand implementation details of a RAG pattern
 - LangChain and WatsonX LLM
 - LangChain, Watson ML and ElasticSearch
 - LangChain, SingleStore
 - LlamaIndex
- Understand chunking/text splitting
- Agentic RAG
- AutoRAG

Section 5: Deployment

This section accounts for approximately 13% of the exam.

5.1. Plan a deployment based on client needs

- Define the deployment lifecycle of a prompt template
- Manage changes to prompts templates in applications
- Understand roles involved with deploying AI Assets
- Understand need around AI Governance
 - Model performance
 - Evaluate inferences
 - Explain outcomes from prompts

5.2. Deploy AI Assets

- Benefits of deployments space
 - Govern lifecycle (dev/evaluation/production)
 - Separate endpoints for applications to call
- Benefits when deploying a prompt template

- Changes to application when using a deployment space

5.3. Deploy a custom model

- Understand requirements of watsonx and foundation model
- Application access to a custom model

5.4. High-level architecture for deployment options

- Version using deployment spaces
- Changes to applications on a prompt version change
- Testing new prompt template version
- Model gateway

5.5. Plan the deployment of prompts for versioning

- Understand the options between managed Software as a Service or Software
- Deployment choices based Architectural pattern
 - RAG
 - Summarization
 - Q&A
- Corpus of Data repository
 - AI Pipelines to update Repository
 - Manage data in corpus
 - Adjusting text chunk size
- Identify endpoint security and stability
- Code generation within PromptLab

Section 6: watsonx - Integration and Model Orchestration

This section accounts for approximately 8% of the exam.

6.1. Integrate watsonx.ai with Other Services

- Connect watsonx.ai to other IBM Cloud services, such as Watson Assistant for chatbot development or Watson Discovery for document understanding, to build end-to-end AI solutions.
 - Identify relevant IBM Cloud services that can be integrated with watsonx.ai (e.g., Watson Assistant, Watson Discovery).
 - Understand the integration points and data exchange mechanisms between watsonx.ai and the chosen services.
 - Configure the integration using watsonx.ai's UI or APIs.
 - Integrate with watsonx.governance
- Leverage watsonx.ai's APIs and SDKs to integrate generative AI models with external
- applications, platforms, or custom workflows, including utilizing LangChain for complex chain creation and management.

- Obtain API keys or credentials for external applications or platforms.
- Write code using watsonx.ai's SDKs to send requests and receive responses from the AI models.
- Handle errors and exceptions during API interactions.

6.2. Orchestrate AI Workflows

- Design and implement workflows utilizing watsonx.ai that combine multiple tasks, including LangChain-based chains, to orchestrate complex generative AI processes.
 - Break down a complex AI task into smaller, manageable steps.
 - Design a workflow using orchestration tools.
- Utilize orchestration tools to automate the execution of workflows, including scheduling, dependency management, and error handling.
 - Schedule workflows to run automatically at specific times or intervals.
 - Implement error handling and retry mechanisms to ensure robustness.
 - Use conditional logic to trigger different workflow branches based on data or model results.
- Monitor and troubleshoot workflow execution, identifying bottlenecks, failures, or performance issues.
 - View workflow logs and execution details within watsonx.ai.
 - Identify and resolve errors or bottlenecks that may occur during workflow execution.
 - Optimize workflow performance by adjusting parameters or parallelizing tasks.
- Agentic RAG

6.3. Understand real-world Integration Scenarios

- Given a business use case, design a comprehensive solution that integrates watsonx.ai with relevant data sources, other IBM Cloud services, and external systems.
 - Identify the necessary data sources, IBM Cloud services, and external systems to be integrated.
 - Design a high-level architecture diagram outlining the data flow and interactions between components.
- Implement the designed solution, leveraging the appropriate watsonx components and tools.
 - Implement the integration using the chosen tools and technologies.
 - Write code to connect data sources, invoke watsonx.ai APIs, and interact with external systems.
 - Test the integration in a development environment.
- Test and validate the integration, ensuring seamless data flow, reliable communication, and expected functionality.
 - Deploy the integrated solution to a production or staging environment.
 - Perform end-to-end testing to verify functionality and data flow.
 - Monitor the solution's performance and address any issues that arise.

- Model Context Protocol

6.4. Develop LLM based applications with LangChain

- Explain the core concepts of LangChain (chains, agents, tools, memory) and their role in building complex conversational AI and generative AI applications.
 - Define the core concepts of LangChain, such as chains, agents, tools, and memory.
 - Explain how LangChain simplifies the development of complex conversational AI and generative AI applications.
 - Describe the benefits of using LangChain, such as modularity, reusability, and extensibility.
- Demonstrate the ability to create and customize LangChain chains, incorporating various components like LLMs, prompt templates, and external data sources.
 - Create a simple LangChain that combines an LLM with a prompt template.
 - Customize a LangChain by adding custom components, such as a tool for interacting with an external API.
- Design and implement LangChain agents that can interact with the environment, make decisions, and take actions based on model outputs and external information.
 - Design a LangChain agent that can perform a specific task, such as answering questions or generating summaries.

Next Steps

1. Take the sample test for [IBM watsonx Generative AI Engineer - Associate](#).
2. Take the assessment test for [IBM watsonx Generative AI Engineer - Associate](#) if you pass the assessment exam, visit <https://home.pearsonvue.com/ibm> to schedule your testing sessions.
3. If you failed the assessment exam, review how you did by section. Focus attention on the sections where you need improvement. Keep in mind that you can take the assessment exam as many times as you would like (\$30 per exam); however, you will still receive the same questions only in a different order.